Anomaly Detection in Network Traffic

A Machine Learning Approach – Deliverable 3

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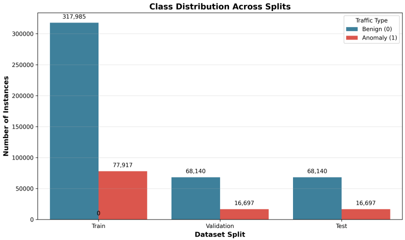
Context and Dataset (CIC-IDS2017)

- **Cybersecurity Challenge:** Identifying malicious traffic dynamically.
- **Dataset Selection:** Canadian Institute for Cybersecurity (CIC-IDS2017).
  - Realistic normal traffic mixed with modern multi-vector attacks (DDoS, Brute Force, Web Attacks, etc.).
- **Computational Hurdles:**
  - Over **2.8 million instances** ($>$ 1 GB raw CSVs) triggering OOM errors on local machines.
  - Severe class imbalance: **80.3% benign** vs. **19.7% anomalous**.

Exploratory Data Analysis (EDA)

Key Insights:

- Highly skewed attributes (Flow Duration: μs to $\geq$ 100s).
- Strong correlations detected between packet volumes and rate features.
- Identified 8 static features (zero-variance) that were pruned.



Native 80/20 Class Distribution

Data Preprocessing Pipeline

1. **Data Cleaning:** Handled column header whitespaces, dropped NaN and Infinite values ($< 0.1\%$ impact).
2. **Feature Selection:** Dropped zero-variance variables (e.g., `Bwd PSH Flags`) to reduce computational noise.
3. **Stratified Sampling:** Extracted a representative 20% sample ($\approx 565,000$ rows) maintaining the 80/20 class split.
4. **Split & Scale:** 70/15/15 train/val/test split. Normalization using *StandardScaler* fitted strictly on training data to prevent **data leakage**.

Evaluation Strategy

- **Imbalance Impact:** Traditional accuracy is misleading (always predicting 'normal' achieves 80.3% accuracy).
- **Selected Metrics:** F1-Score, Recall (critical to catch all attacks), Precision, and AUC-ROC.
- **Multi-Perspective Framework:**
  1. **Original Distribution:** Realistic test environment.
  2. **Balanced Subset (50-50):** Exposes pure algorithm discrimination.
  3. **Per-Class Breakdown:** Recall analysis across different attack vectors.

State of the Art in Network Anomaly Detection

Table: Summary of literature models and findings.

Method	Type	Data Benchmark	AUC
Isolation Forest (2008)	Tree Ensemble	KDD'99	0.96
LOF (2000)	Density Ratio	Synthetic	0.89
One-Class SVM (2001)	Kernel Boundary	USPS	0.87
Autoencoder (2014)	Deep learning	MSDS	0.93
Kitsune (2018)	Ensemble AE	Real Traffic	0.99
DAGMM (2018)	Deep Hybrid	KDD'99	0.98

Major Conclusions from Review:

- Tree ensembles and neural networks consistently outperform density methods in dimensions > 20.
- Hyperparameter tuning is rarely reported systematically in cybersecurity papers.

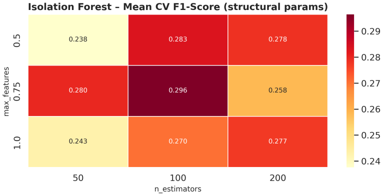
Classical Algorithms Evaluated

- **Isolation Forest:** Builds an ensemble of isolation trees. Anomalies are isolated closer to the root of the trees.
- **One-Class SVM:** Constructs a maximum-margin hyperplane in a high-dimensional kernel space to separate normal points.
- **Local Outlier Factor (LOF):** Compares the local reachability density of a point against its k-nearest neighbors.

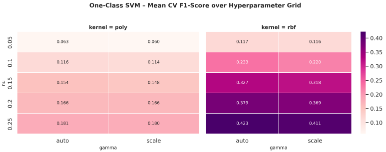
Hyperparameter tuning strategy:

- **3-Fold Stratified Cross-Validation (CV)** conducted on the training split to select optimal parameters.
- Fixed held-out test set preserved strictly for final comparisons.

Classical Models Tuning Heatmaps



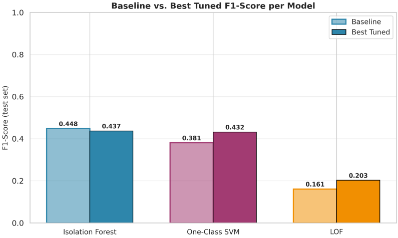
Isolation Forest CV F1-Score



One-Class SVM (RBF vs. Poly) CV F1

- **Isolation Forest:** Contamination is the most sensitive parameter.
- **One-Class SVM:** RBF kernel consistently beats Polynomial.

Baseline vs. Tuned Performance



Test F1 Comparison (Baseline vs. Best)

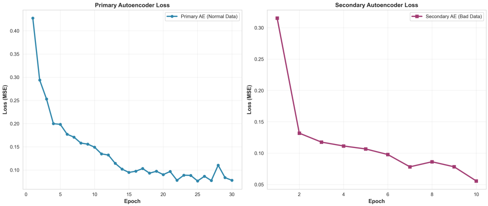
Results of Tuning:

- **Isolation Forest:** Significant boost from tuning (+5.6% F1 score).
- **One-Class SVM:** Moderate improvement (+3.0% F1).
- **LOF:** Negligible change. Confirming that LOF's density ratios break down in high-dimensional (70D) spaces.

Deep Learning Architectures (Autoencoders)

- **Primary Autoencoder (Normal-Trained):**
  - Trained exclusively on benign network traffic ($y = 0$).
  - Reconstruction MSE loss: $\mathcal{L}_{\text{MSE}} = \frac{1}{N} \sum \|\mathbf{x} - \hat{\mathbf{x}}\|^2$.
  - Anomaly Score: $s_{\text{primary}}(\mathbf{x}) = \|\mathbf{x} - \hat{\mathbf{x}}\|^2$.
- **Secondary Autoencoder (Anomaly-Trained):**
  - Trained exclusively on malicious traffic ($y = 1$).
  - Learns statistical similarities of attacks.
  - Anomaly Score: $s_{\text{secondary}}(\mathbf{x}) = -\|\mathbf{x} - \hat{\mathbf{x}}_2\|^2$.
- **Ensemble Autoencoder:**
  - Logical OR-gate decision framework: $\hat{y}_{\text{ensemble}} = \hat{y}_{\text{primary}} \lor \hat{y}_{\text{secondary}}$.
  - Maximizes detection rates, minimizing dangerous false negatives.

Autoencoders Training Convergence



Training and validation Loss curves for both network branches

Convergence Analysis:

- Dynamic training using early stopping on validation reconstruction loss.
- Primary AE converged and stopped at epoch 25.
- Secondary AE converged and stopped at epoch 40.
- No overfitting observed on validation set.

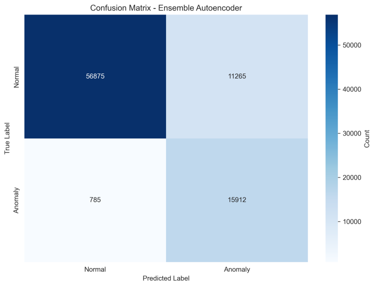
Final Metric Comparison (Test Set)

Table: Model results under original and balanced test splits.

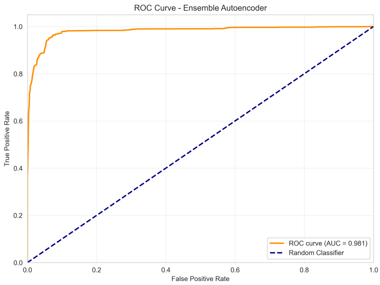
	Original (Imbalanced)			Balanced (50-50)		
Model	**Precision**	**Recall**	**F1**	**Precision**	**Recall**	**F1**
Isolation Forest	0.4351	0.4453	0.4402	0.7639	0.4453	0.5627
One-Class SVM	0.3792	0.3859	0.3825	0.7202	0.3859	0.5026
LOF	0.4417	0.6678	0.5317	0.7680	0.6678	0.7144
Autoencoder	0.5308	0.5476	0.5391	0.8234	0.5476	0.6578
Second AE	**0.7615**	0.7812	**0.7712**	**0.9265**	0.7812	0.8477
Ensemble AE	0.5722	**0.9710**	0.7200	0.8448	**0.9710**	**0.9035**

- **Deep Learning Dominance:** PyTorch Autoencoders far exceed classical.
- **Ensemble AE:** Best recall (**97.1%**) and AUC-ROC (**0.947**), crucial for maximum security.
- **Second AE:** Best precision (**76.15%** original) and F1 (**0.7712**), minimizing false alarms.

Ensemble Diagnostics



Confusion Matrix



ROC Curve (AUC = 0.947)

- Out of 16,697 actual anomalies in the test set, only 484 are missed.
- The 57.22% precision is due to the logical OR aggregating false positives, which is an acceptable compromise for security operations.

Overfitting and Underfitting Analysis

- **Underfitting**:
  - *Local Outlier Factor (LOF):* Underfits severely in 70D because distance concentration hides outlier patterns.
  - *Baselines:* Isolation Forest and OC-SVM default parameters fail to capture multi-vector attack profiles.
- **Overfitting**:
  - *One-Class SVM:* Highly prone to overfitting if γ is too high, leading to a tight envelope and massive false alarms.
  - *Autoencoders:* Prevented from overfitting (identity mapping) by a compact bottleneck architecture ($70 \rightarrow 8$).
- **Regularization:** Managed by monitoring train/val loss curves.

Critical Reflection and Responsible AI

- **Theory vs. Practice:** In high dimensions, density concepts (like LOF) degrade. Modeling must align with data geometry.
- **Ethical Dimensions:**
  - False positives can block legitimate users, impacting business.
  - Models are trained on university network traffic from 2017; generalizability to modern, zero-day corporate networks requires validation.
- **AI Tools Disclosure:** Used Gemini Pro for debugging PyTorch tensors and Scite.ai/Elicit.com for literature reference verification.

Conclusions and Future Work

- **Key Takeaways:**
  1. Unsupervised Deep Learning is highly successful for anomaly-based IDS.
  2. Ensemble AE provides maximal security monitoring (97.1% recall).
  3. Secondary AE minimizes false alarms, suitable for automated mitigation.
- **Future Work:**
  - Integrating Deep Autoencoding Gaussian Mixture Models (DAGMM).
  - Adding Attention layers inside the encoder/decoder stacks.
  - Implementing weighted scoring logic instead of strict binary OR.

Thank you. Questions?